

ADAPTIVE SOFTWARE ENGINEERING(24I3201)

ASSETFLOW – ENTERPRISE IT ASSET MANAGEMENT PLATFORM WITH INTELLIGENT ASSET TRACKING & LIFECYCLE MONITORING

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ABSTRACT

AssetFlow is a centralized web-based system designed to simplify and automate IT asset and inventory management. It helps organizations efficiently track devices such as laptops, desktops, monitors, printers, networking equipment, and accessories.

The system allows authorized users to add, update, categorize, assign, transfer, and monitor assets. It maintains details such as asset ID, category, purchase information, employee, location, condition, availability, and status. A dashboard provides insights into total, available, assigned, and maintenance assets.

The frontend uses React.js, while Spring Boot 3 provides RESTful APIs and backend services. PostgreSQL stores asset and user data securely. Docker supports deployment, while GitHub and Jira enable version control, collaboration, and Agile project management.

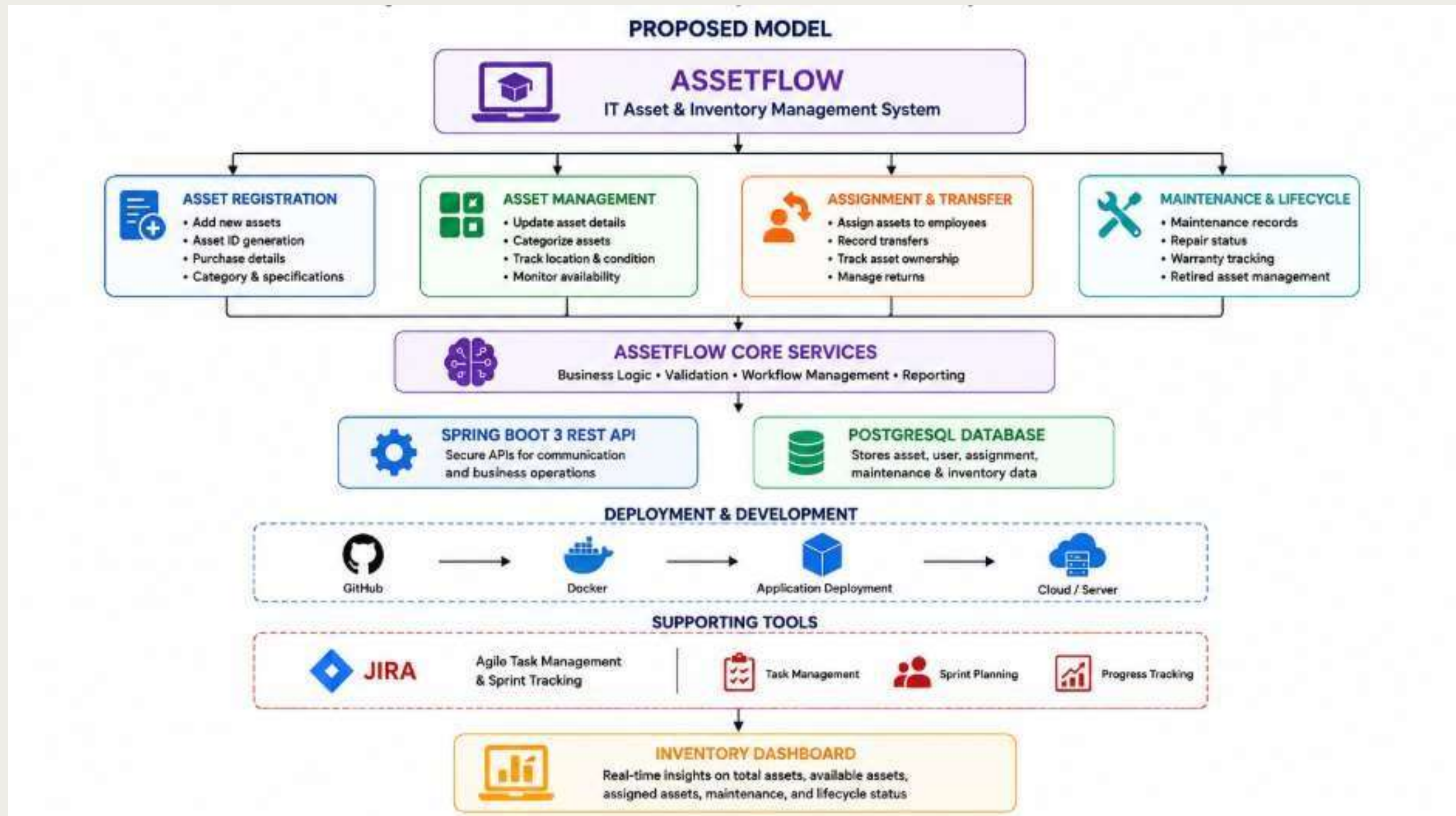
Existing Model

Existing System	Main Features	Limitation / Gap
Snipe-IT	Asset tracking, assignments, licenses, locations, and inventory management	Mainly focused on asset tracking with limited workflow automation
ManageEngine AssetExplorer	IT asset tracking, lifecycle management, software and hardware inventory	Complex for small organizations and requires extensive configuration
ServiceNow ITAM	Asset lifecycle, service management, procurement, and tracking	Expensive and more suitable for large enterprises
Asset Panda	Asset tracking, barcode scanning, custom workflows, and reporting	Customization can be complex and costly
Spreadsheet-Based Systems	Basic asset records, employee assignments, and inventory tracking	Manual updates, duplicate data, data inconsistency, and poor real-time tracking

PROPOSED MODEL

AssetFlow integrates asset registration, inventory tracking, assignment, transfer, maintenance, and lifecycle monitoring into a single centralized system.

Admin/User → Asset Registration → Asset Categorization → Asset Assignment → Asset Tracking → Maintenance Monitoring → Asset Return/Transfer → Lifecycle Status → Inventory Dashboard



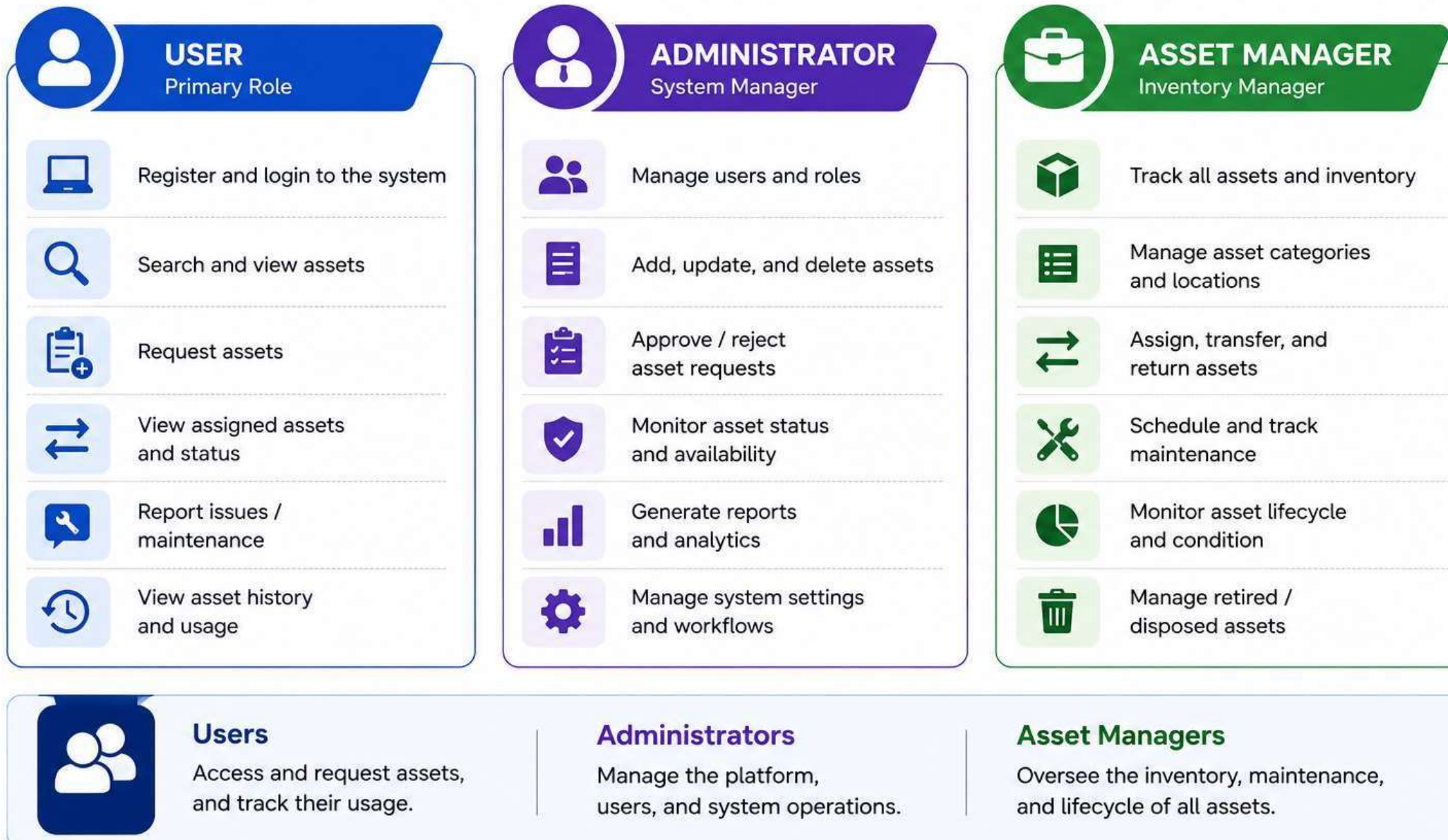
Users

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USERS

AssetFlow supports multiple user roles with distinct permissions and functionalities.



Methodology

Requirement Analysis



Agile / Scrum Planning



System Design & Database Design



Frontend & Backend Development



Asset Management Module Integration



Testing & Validation



Docker Deployment & GitHub Integration

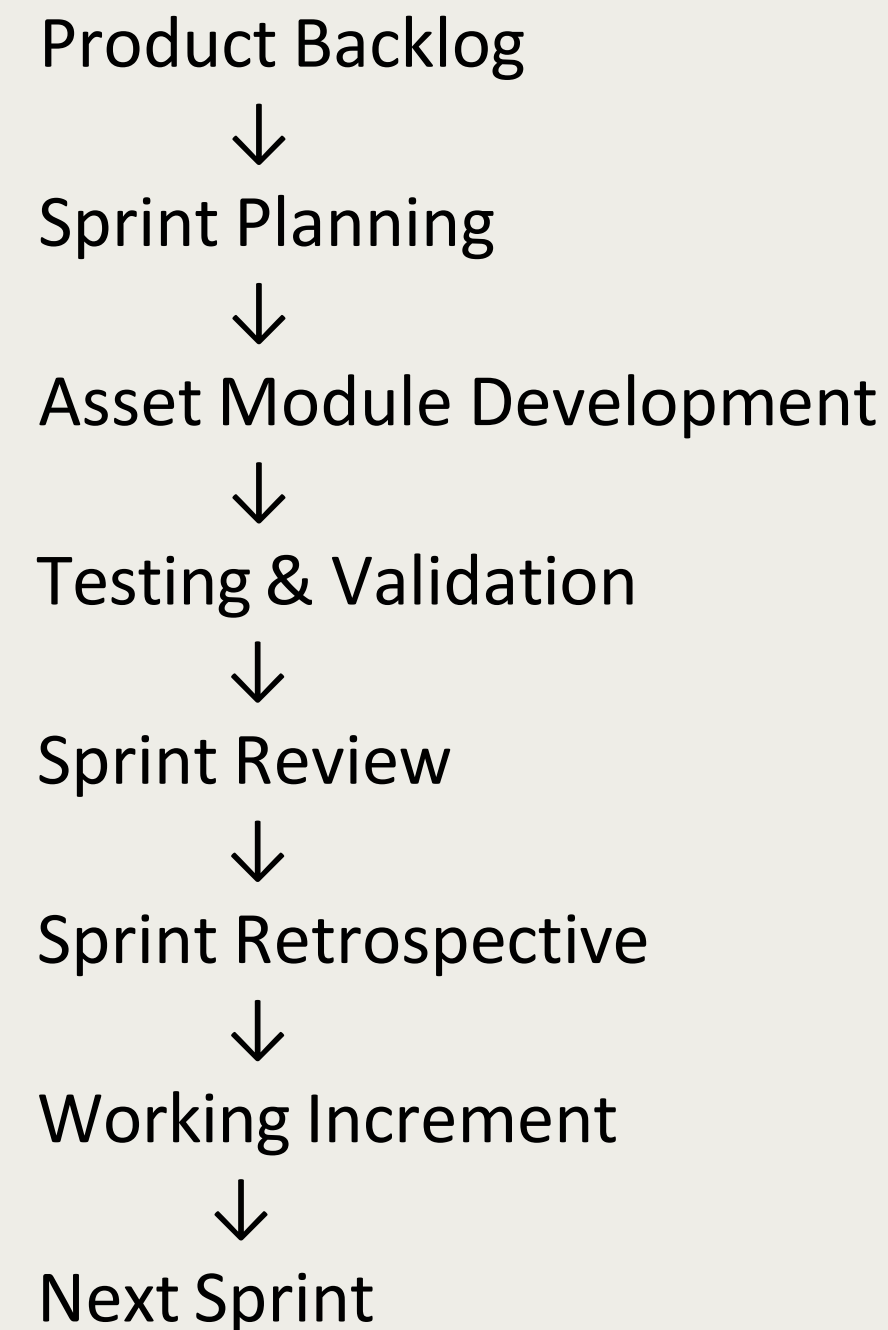


Monitoring & Continuous Improvement

Agile Scrum Model

Why Agile Scrum?

- **Iterative development** – Build AssetFlow in small functional modules.
- **Continuous feedback** – Get regular feedback from users and team members.
- **Easy adaptation** – Quickly adapt to changing asset-management requirements.
- **Frequent testing** – Test and improve each module after every sprint.
- **Team collaboration** – Supports coordination among developers and stakeholders.



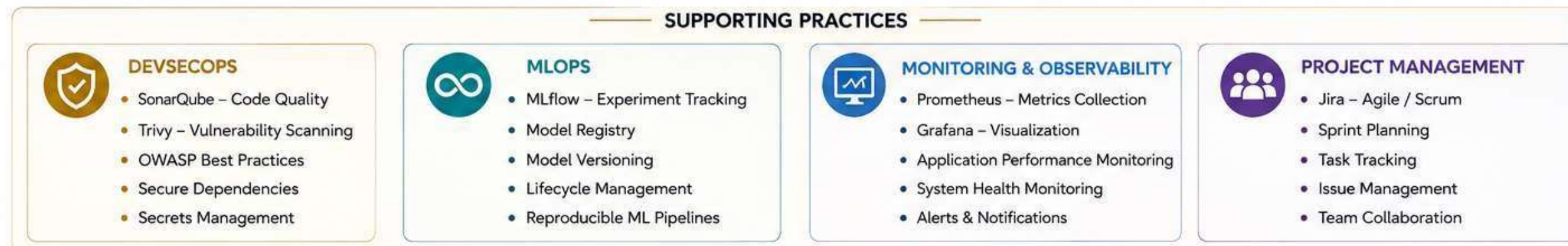
Technical Aspects

TECHNICAL ASPECTS

AssetFlow is built using modern technologies and best practices to ensure secure, scalable and efficient IT Asset & Inventory Management.



SUPPORTING PRACTICES



SYSTEM WORKFLOW



CONCLUSION

- AssetFlow provides a centralized and efficient solution for managing an organization's IT assets throughout their complete lifecycle. By combining React.js, Spring Boot, PostgreSQL, Docker, GitHub, and Jira, the system enables secure asset tracking, assignment, maintenance, and inventory management.
- It reduces manual effort, improves data accuracy, and provides better visibility into asset availability and usage. The modular and scalable architecture also allows AssetFlow to be extended with additional features in the future.
- AssetFlow ultimately helps organizations achieve smarter asset utilization, improved accountability, and more efficient IT operations.

THANK YOU